Tugas Akhir Analitika **Data Manajemen (Coding)** *(North)*

Presented By:
Kelompok 1 & 10

Presented To:
Nurmala, S.E., M.M., Ph.D.

Iman Satyo Adi
2306208262

Daffa Adli
2306206433

Arsha Putra S
230611548

Carmen Kiara
2306223856

Habel Nababan
2306209181

Darryl A. R.
220622123

04

Coding

02. Predictive Analysis | Clustering

Clustering adalah teknik yang digunakan untuk mengelompokkan titik-titik data ke dalam beberapa kelompok (cluster) sehingga data dalam satu kelompok memiliki karakteristik yang lebih mirip satu sama lain dibandingkan dengan data di kelompok lain.
Dalam Phyton, library yang paling sering digunakan adalah scikit learn atau sklearn. K dalam library tersebut menandakan clustering.

0.1 Importing Libraries

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

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02. Predictive Analysis | Clustering

0.2 Menentukan ID Serta Variabel Dependen dan Independen

```
# 1. Aggregate data by Customer ID
customer_data = df.groupby('Customer ID').agg({
    'Sales': 'sum',
    'Profit': 'sum'
}).reset_index()

# 2. Scale the features (K-Means is sensitive to scale)
X_cluster = customer_data[['Sales', 'Profit']]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_cluster)
```

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02. Predictive Analysis | Clustering

0.3 Menentukan Cluster dan Visualisasi Data

```
# 3. Apply K-Means (Let's choose 3 clusters for High, Medium, Low value customers)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
customer_data['Cluster'] = kmeans.fit_predict(X_scaled)

# 4. Visualize the clusters
plt.figure(figsize=(10, 6))
sns.scatterplot(data=customer_data, x='Sales', y='Profit', hue='Cluster', palette='viridis')
plt.title('Customer Segments based on Sales and Profit')
plt.xlabel('Total Sales')
plt.ylabel('Total Profit')
plt.show()

# See the average stats per cluster
display(customer_data.groupby('Cluster')[['Sales', 'Profit']].mean())
```

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02. Predictive Analysis | Clustering

Cluster	Sales	Profit
0	838.644135	57.809145
1	2771.417391	523.239130
2	6218.033333	1458.033333

Berdasarkan grafik tersebut, terdapat 3 segmentasi pasar. Segmentasi **cluster yang berwarna kuning (2)** merupakan cluster yang terdiri dari sales terbanyak dan menghasilkan profitabitabilitas tertinggi **(high value)**. Kemudian terdapat segmentasi **cluster yang berwarna hijau (1)** yang memberikan kontribusi lebih rendah secara sales dan juga angka profitabilitas. Sedangkan, segmentasi **cluster berwarna ungu (0)** memiliki sales dan profitabilitas yang jomplang sehingga menempatkan cluster tersebut menjadi yang paling rendah.

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02. Predictive Analysis | Association

01. Importing Libraries

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
from itertools import combinations
from collections import Counter

# Plot styling
plt.rcParams['figure.dpi'] = 120
plt.rcParams['font.family'] = 'DejaVu Sans'
sns.set_style('whitegrid')
```

02. Load & Explore Data

```python
df = pd.read_csv('cleaned_north_superstore-3.csv')

print(f'Dataset shape: {df.shape}')
print(f'Unique Orders: {df["Order ID"].nunique()}')
print(f'Unique Sub-Categories: {df["Sub-Category"].nunique()}')
print(f'Date range: {df["Order Date"].min()} → {df["Order Date"].max()}')
df.head()

# Sub-category frequency
fig, ax = plt.subplots(figsize=(12, 5))
counts = df['Sub-Category'].value_counts()
bars = ax.barh(counts.index, counts.values, color=sns.color_palette('Blues_r', len(counts)))
ax.set_xlabel('Number of Order Lines', fontsize=12)
ax.set_title('Sub-Category Frequency in Dataset', fontsize=14, fontweight='bold')
for bar, val in zip(bars, counts.values):
    ax.text(val + 5, bar.get_y() + bar.get_height()/2, str(val), va='center', fontsize=9)
plt.tight_layout()
plt.show()
```

02. Predictive Analysis | Association

03. Build Transaction Baskets

```
# Group each order into a set of unique sub-categories
basket = df.groupby('Order ID')['Sub-Category'].apply(set)
total_orders = len(basket)

# Only keep multi-item orders for pair analysis
multi_item_orders = basket[basket.apply(len) >= 2]

print(f'Total orders: {total_orders}')
print(f'Single-item orders: {total_orders - len(multi_item_orders)}')
print(f'Multi-item orders: {len(multi_item_orders)} ({len(multi_item_orders)/total_orders*100:.1f}%)')

# Basket size distribution
basket_sizes = basket.apply(len)
fig, ax = plt.subplots(figsize=(8, 4))
size_counts = basket_sizes.value_counts().sort_index()
ax.bar(size_counts.index, size_counts.values, color='steelblue', edgecolor='white', linewidth=0.8)
ax.set_xlabel('Number of Unique Sub-Categories per Order')
ax.set_ylabel('Number of Orders')
ax.set_title('Basket Size Distribution', fontweight='bold')
for x, y in zip(size_counts.index, size_counts.values):
    ax.text(x, y + 5, str(y), ha='center', fontsize=9)
plt.tight_layout()
plt.show()
```

02. Predictive Analysis | Association

04. Compute Support, Confidence, and Lift

```python
# Count individual item frequencies
item_counts = Counter()
for items in basket:
    for item in items:
        item_counts[item] += 1

# Count pair frequencies
pair_counts = Counter()
for items in multi_item_orders:
    for a, b in combinations(sorted(items), 2):
        pair_counts[(a, b)] += 1

# Build association rules table
rules = []
for (a, b), cnt in pair_counts.items():
    support     = cnt / total_orders
    conf_ab     = cnt / item_counts[a]    # P(B|A)
    conf_ba     = cnt / item_counts[b]    # P(A|B)
    lift_ab     = conf_ab / (item_counts[b] / total_orders)
    lift_ba     = conf_ba / (item_counts[a] / total_orders)
```

```python
    rules.append({'Antecedent': a, 'Consequent': b,
                  'Support': round(support, 4),
                  'Confidence': round(conf_ab, 4),
                  'Lift': round(lift_ab, 4),
                  'Count': cnt})
    rules.append({'Antecedent': b, 'Consequent': a,
                  'Support': round(support, 4),
                  'Confidence': round(conf_ba, 4),
                  'Lift': round(lift_ba, 4),
                  'Count': cnt})

rules_df = pd.DataFrame(rules).sort_values('Lift', ascending=False).reset_index(drop=True)

print(f'Total rules generated: {len(rules_df)}')
print(f'Rules with Lift > 1.2: {len(rules_df[rules_df["Lift"] > 1.2])}')
print(f'Lift range: {rules_df["Lift"].min():.3f} – {rules_df["Lift"].max():.3f}')
rules_df.head(20)
```

02. Predictive Analysis | Association

05. Filter Strong Rules

```python
# Strong rules: Lift > 1.1 and minimum 10 co-occurrences
strong_rules = rules_df[
    (rules_df['Lift'] > 1.1) &
    (rules_df['Count'] >= 10)
].copy().reset_index(drop=True)

print(f'Strong rules: {len(strong_rules)}')
strong_rules[['Antecedent','Consequent','Support','Confidence','Lift','Count']]
```

	Antecedent	Consequent	Support	Confidence	Lift	Count
0	Copiers	Tables	0.0047	0.0442	1.5534	11
1	Tables	Copiers	0.0047	0.1642	1.5534	11
2	Tables	Labels	0.0042	0.1493	1.5027	10
3	Labels	Tables	0.0042	0.0427	1.5027	10
4	Furnishings	Accessories	0.0149	0.1423	1.4386	35
5	Accessories	Furnishings	0.0149	0.1502	1.4386	35
6	Envelopes	Supplies	0.0157	0.1370	1.4349	37
7	Supplies	Envelopes	0.0157	0.1644	1.4349	37
8	Envelopes	Accessories	0.0153	0.1333	1.3482	36
9	Accessories	Envelopes	0.0153	0.1545	1.3482	36
10	Paper	Envelopes	0.0140	0.1535	1.3393	33
11	Envelopes	Paper	0.0140	0.1222	1.3393	33
12	Machines	Binders	0.0136	0.2602	1.2823	32
13	Binders	Machines	0.0136	0.0669	1.2823	32
14	Accessories	Machines	0.0064	0.0644	1.2331	15

02. Predictive Analysis | Association

06. Support vs Confidence (Scatter)

```python
fig, ax = plt.subplots(figsize=(11, 7))

# All rules in background
scatter_all = ax.scatter(
    rules_df['Support'], rules_df['Confidence'],
    c=rules_df['Lift'], cmap='YlOrRd',
    alpha=0.4, s=40, zorder=2
)

# Highlight strong rules
scatter_strong = ax.scatter(
    strong_rules['Support'], strong_rules['Confidence'],
    c=strong_rules['Lift'], cmap='YlOrRd',
    alpha=1.0, s=100, edgecolors='black', linewidth=0.8, zorder=3
)

# Label top 8 by lift
top8 = strong_rules.head(8)
for _, row in top8.iterrows():
    label = f"{row['Antecedent']}\n→ {row['Consequent']}"
    ax.annotate(label,
                xy=(row['Support'], row['Confidence']),
                xytext=(8, 4), textcoords='offset points',
                fontsize=7.5, color='#222222',
                bbox=dict(boxstyle='round,pad=0.2', fc='white', alpha=0.7, ec='gray'))
```

```python
cbar = plt.colorbar(scatter_all, ax=ax)
cbar.set_label('Lift', fontsize=11)

ax.set_xlabel('Support  (fraction of all orders containing this pair)', fontsize=11)
ax.set_ylabel('Confidence  (P(Consequent | Antecedent))', fontsize=11)
ax.set_title('Association Rules: Support vs Confidence\n(Darker = Higher Lift | Outlined = Strong Rules)',
             fontsize=13, fontweight='bold')

# Reference lines
ax.axhline(0.10, color='gray', linestyle='--', linewidth=0.8, alpha=0.5)
ax.axvline(0.01, color='gray', linestyle='--', linewidth=0.8, alpha=0.5)

plt.tight_layout()
plt.savefig('plot1_scatter.png', bbox_inches='tight')
plt.show()
print('Saved: plot1_scatter.png')
```

02. Predictive Analysis | Association

07. Top Rules by Lift (Bar Chart)

```
top15 = strong_rules.head(15).copy()
top15['Rule'] = top15['Antecedent'] + '  →  ' + top15['Consequent']
top15 = top15.sort_values('Lift')

fig, ax = plt.subplots(figsize=(12, 7))

colors = plt.cm.RdYlGn(np.linspace(0.3, 0.9, len(top15)))
bars = ax.barh(top15['Rule'], top15['Lift'], color=colors, edgecolor='white', height=0.6)

# Value labels
for bar, val, conf, cnt in zip(bars, top15['Lift'], top15['Confidence'], top15['Count']):
    ax.text(bar.get_width() + 0.01, bar.get_y() + bar.get_height()/2,
            f'Lift={val:.2f}  Conf={conf:.2f}  n={cnt}',
            va='center', fontsize=8.5)

ax.axvline(1.0, color='red', linestyle='--', linewidth=1.2, label='Lift = 1.0 (random)')
ax.set_xlabel('Lift', fontsize=12)
ax.set_title('Top 15 Association Rules by Lift', fontsize=14, fontweight='bold')
ax.legend(fontsize=10)
ax.set_xlim(0, top15['Lift'].max() + 0.35)

plt.tight_layout()
plt.savefig('plot2_bar.png', bbox_inches='tight')
plt.show()
print('Saved: plot2_bar.png')
```

02. Predictive Analysis | Association

08. Co-occurence Heatmap (Raw Counts)

```
subcats = sorted(df['Sub-Category'].unique())

# Build co-occurrence matrix with RAW counts
co_matrix = pd.DataFrame(0, index=subcats, columns=subcats)
for items in basket:
    for a, b in combinations(sorted(items), 2):
        co_matrix.loc[a, b] += 1
        co_matrix.loc[b, a] += 1

# Mask the diagonal (self-pairs are meaningless)
mask = np.eye(len(subcats), dtype=bool)

fig, ax = plt.subplots(figsize=(14, 12))
sns.heatmap(
    co_matrix,
    mask=mask,
    annot=True,
    fmt='d',                # Integer counts — not 0/1
    cmap='Blues',
    linewidths=0.4,
    linecolor='white',
    ax=ax,
    cbar_kws={'label': 'Number of Orders with Both Items'}
)
```

```
ax.set_title('Sub-Category Co-occurrence Heatmap\n(Raw Order Counts)',
             fontsize=14, fontweight='bold', pad=15)
ax.set_xticklabels(ax.get_xticklabels(), rotation=45,
                   ha='right', fontsize=9)
ax.set_yticklabels(ax.get_yticklabels(), rotation=0, fontsize=9)

plt.tight_layout()
plt.savefig('plot3_heatmap.png', bbox_inches='tight')
plt.show()
print('Saved: plot3_heatmap.png')
```

02. Predictive Analysis | Association

09. Lift Heatmap (Relationship Strength)

```python
# Build lift matrix
lift_matrix = pd.DataFrame(np.nan, index=subcats, columns=subcats)

for (a, b), cnt in pair_counts.items():
    support   = cnt / total_orders
    p_a       = item_counts[a] / total_orders
    p_b       = item_counts[b] / total_orders
    lift      = support / (p_a * p_b)
    lift_matrix.loc[a, b] = round(lift, 2)
    lift_matrix.loc[b, a] = round(lift, 2)

# Fill diagonal with 1 (or NaN to hide)
np.fill_diagonal(lift_matrix.values, np.nan)

fig, ax = plt.subplots(figsize=(14, 12))
sns.heatmap(
    lift_matrix,
    annot=True,
    fmt='.2f',
    cmap='RdYlGn',    # Red < 1 (negative), Green > 1 (positive)
    center=1.0,
    linewidths=0.4,
    linecolor='white',
    ax=ax,
    vmin=0.7, vmax=1.6,
    cbar_kws={'label': 'Lift (>1 = positive association)'}
)
```

```python
ax.set_title('Association Lift Matrix\n(Green = Positively associated | Red = Negatively associated)',
             fontsize=14, fontweight='bold', pad=15)
ax.set_xticklabels(ax.get_xticklabels(), rotation=45, ha='right', fontsize=9)
ax.set_yticklabels(ax.get_yticklabels(), rotation=0, fontsize=9)

plt.tight_layout()
plt.savefig('plot4_lift_heatmap.png', bbox_inches='tight')
plt.show()
print('Saved: plot4_lift_heatmap.png')
```

02. Predictive Analysis | Association

10. Bubble Chart (Support × Confidence × Lift)

```python
plot_df = strong_rules.copy()
plot_df['Rule'] = plot_df['Antecedent'] + ' → ' + plot_df['Consequent']

fig, ax = plt.subplots(figsize=(12, 7))

scatter = ax.scatter(
    plot_df['Support'],
    plot_df['Confidence'],
    s=plot_df['Lift'] ** 4 * 200,   # Bubble size = lift
    c=plot_df['Count'],
    cmap='plasma',
    alpha=0.75,
    edgecolors='white',
    linewidth=0.8
)

cbar = plt.colorbar(scatter, ax=ax)
cbar.set_label('Order Count (co-occurrences)', fontsize=10)

for _, row in plot_df.iterrows():
    ax.annotate(
        row['Rule'],
        xy=(row['Support'], row['Confidence']),
        xytext=(5, 5), textcoords='offset points',
        fontsize=7.5,
        bbox=dict(boxstyle='round,pad=0.2', fc='white', alpha=0.6)
    )
```

```python
ax.set_xlabel('Support', fontsize=12)
ax.set_ylabel('Confidence', fontsize=12)
ax.set_title('Strong Association Rules\n(Bubble size = Lift, Color = Order Count)',
             fontsize=13, fontweight='bold')

# Legend for bubble size
for lift_val in [1.1, 1.2, 1.4]:
    ax.scatter([], [], s=lift_val**4*200, c='gray', alpha=0.5, label=f'Lift = {lift_val}')
ax.legend(title='Bubble Size', framealpha=0.8, fontsize=9)

plt.tight_layout()
plt.savefig('plot5_bubble.png', bbox_inches='tight')
plt.show()
print('Saved: plot5_bubble.png')
```

02. Predictive Analysis | Association

Filtering Strong Rules

Strong rules: 42

	Antecedent	Consequent	Support	Confidence	Lift	Count
0	Copiers	Tables	0.0047	0.0442	1.5534	11
1	Tables	Copiers	0.0047	0.1642	1.5534	11
2	Tables	Labels	0.0042	0.1493	1.5027	10
3	Labels	Tables	0.0042	0.0427	1.5027	10
4	Furnishings	Accessories	0.0149	0.1423	1.4386	35
5	Accessories	Furnishings	0.0149	0.1502	1.4386	35
6	Envelopes	Supplies	0.0157	0.1370	1.4349	37
7	Supplies	Envelopes	0.0157	0.1644	1.4349	37
8	Envelopes	Accessories	0.0153	0.1333	1.3482	36
9	Accessories	Envelopes	0.0153	0.1545	1.3482	36
10	Paper	Envelopes	0.0140	0.1535	1.3393	33
11	Envelopes	Paper	0.0140	0.1222	1.3393	33
12	Machines	Binders	0.0136	0.2602	1.2823	32
13	Binders	Machines	0.0136	0.0669	1.2823	32
14	Accessories	Machines	0.0064	0.0644	1.2331	15
15	Machines	Accessories	0.0064	0.1220	1.2331	15
16	Labels	Copiers	0.0127	0.1282	1.2131	30
17	Copiers	Labels	0.0127	0.1205	1.2131	30
18	Paper	Bookcases	0.0119	0.1302	1.2128	28

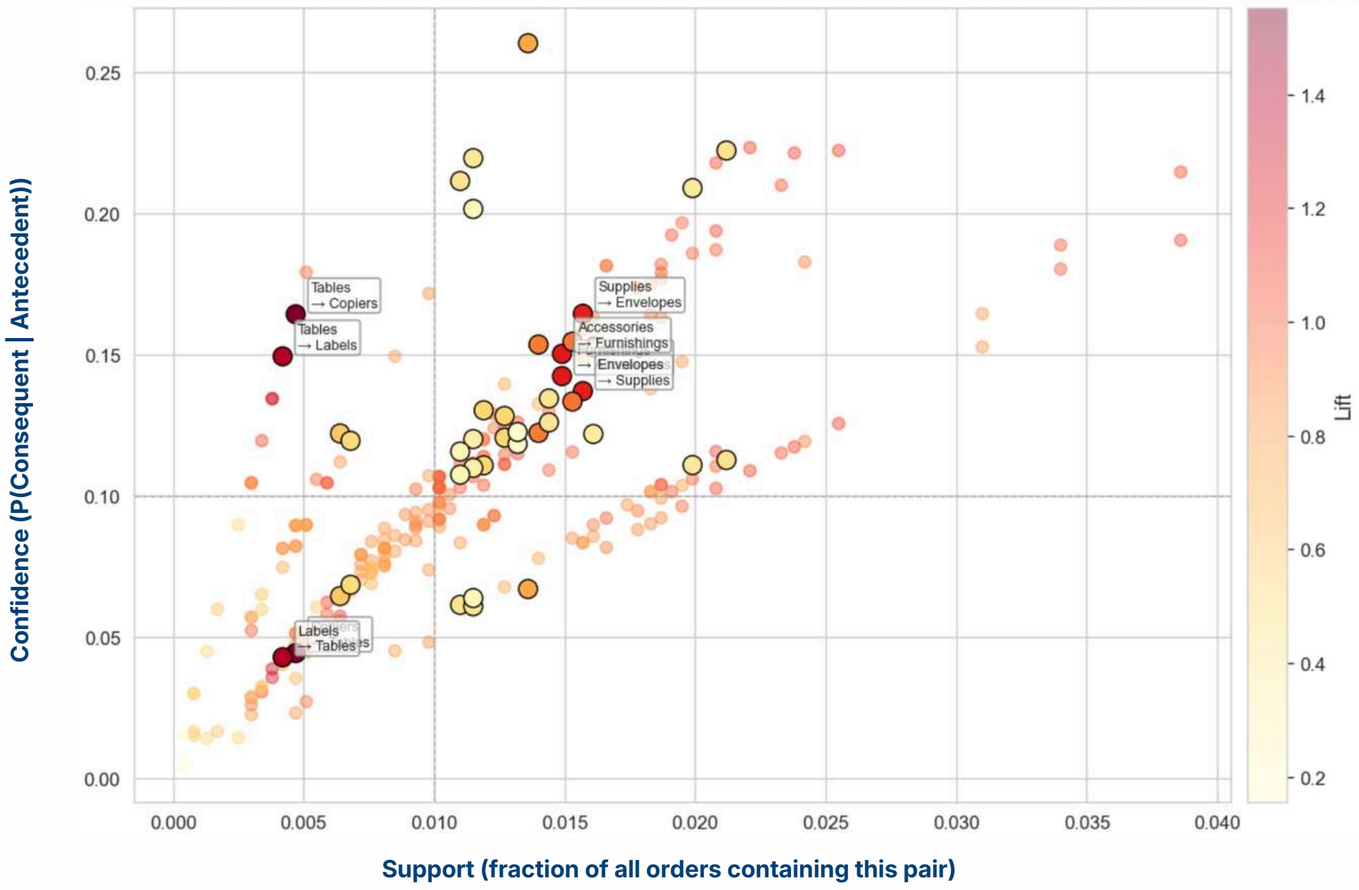
Ditemukan **42 kombinasi produk** yang paling sering dibeli secara bersamaan dengan korelasi yang terbukti kuat secara statistik.

Tiga Metrik:

1. **Support:** Persentase frekuensi kombinasi produk muncul dari total seluruh transaksi.
2. **Confidence:** Tingkat probabilitas (Contoh: 26% pelanggan yang membeli Machines pasti membeli Binders).
3. **Lift (> 1):** Kekuatan asosiasi. Semakin tinggi dari angka 1, semakin membuktikan kombinasi tersebut adalah kebutuhan riil pelanggan.

02. Predictive Analysis | Association

Association Rules: Support vs Confidence



Darker = Higher Lift
Outlined = Strong Rules

Temuan:

Tables → Copiers: Memiliki nilai Lift tertinggi (titik merah paling pekat). Walau jarang terjadi di total transaksi (Support <0.005), jika pelanggan membeli Tables, probabilitas mereka menambah Copiers cukup tinggi (Confidence ~0.16).

Supplies → Envelopes: Kombinasi yang sangat ideal. Lebih sering dibeli bersamaan (Support ~0.017) dan memiliki ikatan korelasi yang kuat dan pasti (Confidence ~0.17, merah pekat).

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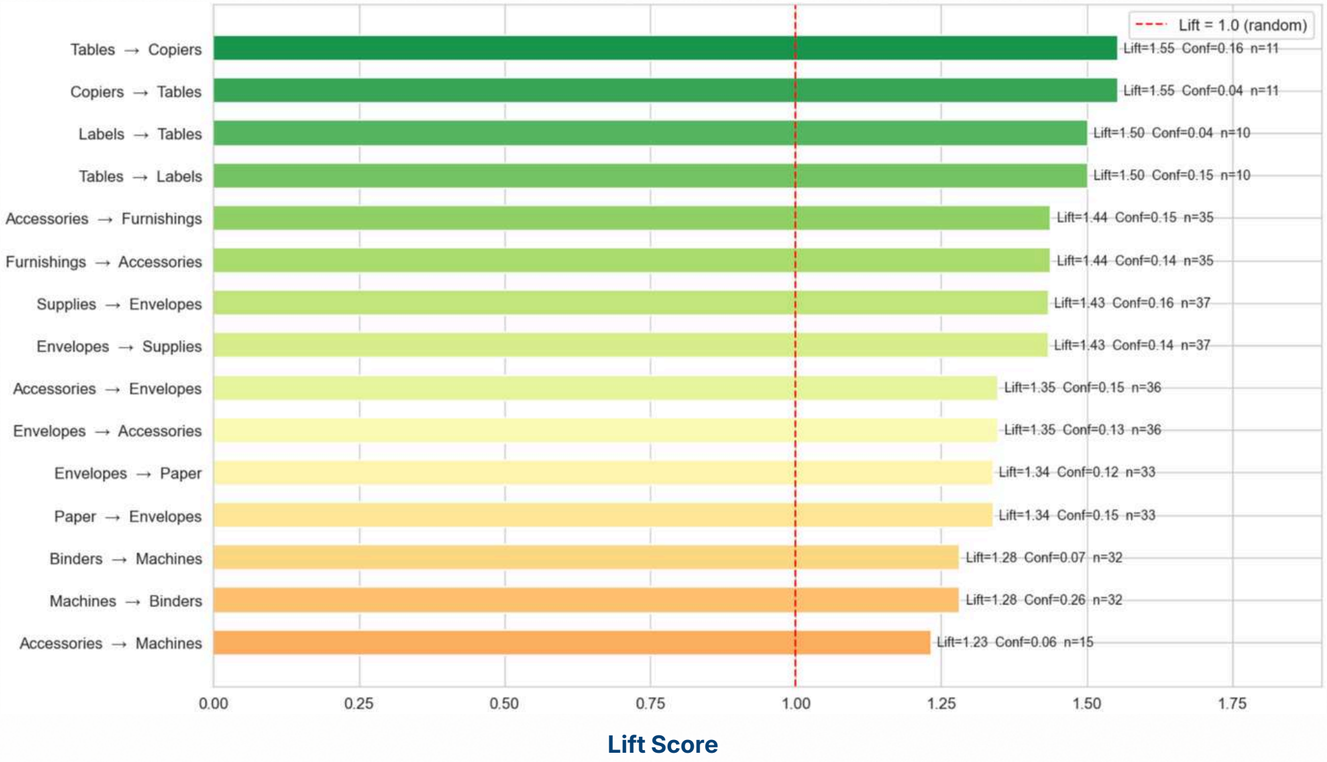
Iman Satyo Adi (2306208262)

02. Predictive Analysis | Association

Association Rules by Lift



Garis putus-putus merah adalah batas Lift = 1.0 (pembelian acak). Semua kombinasi di grafik ini memiliki nilai di atas 1.0, yang berarti semuanya punya korelasi positif yang signifikan (saling mendorong pembelian).

Grafik ini mengurutkan 15 kombinasi produk berdasarkan kekuatan hubungan (Lift).

Korelasi Terkuat (Peringkat 1 & 2): Pasangan Tables ↔ Copiers memiliki ikatan paling kuat dengan nilai Lift = 1.55. Jika pelanggan membeli Tables, probabilitas mereka juga membeli Copiers mencapai 16% (Conf=0.16), meskipun sejauh ini baru terjadi di 11 transaksi (n=11).

Kepastian Tertinggi (Highest Confidence): Pada Machines → Binders. Walau Lift-nya berada di tengah (1.28), probabilitasnya adalah yang paling tinggi di grafik ini, yaitu 26% (Conf=0.26) dari 32 transaksi. Artinya, lebih dari seperempat pembeli Machines pasti mencari Binders.

Kombinasi Kuat & Cukup Populer: Supplies ↔ Envelopes tidak hanya punya korelasi kuat (Lift = 1.43), tapi juga frekuensi transaksi yang cukup tinggi di daftar ini (n=37).

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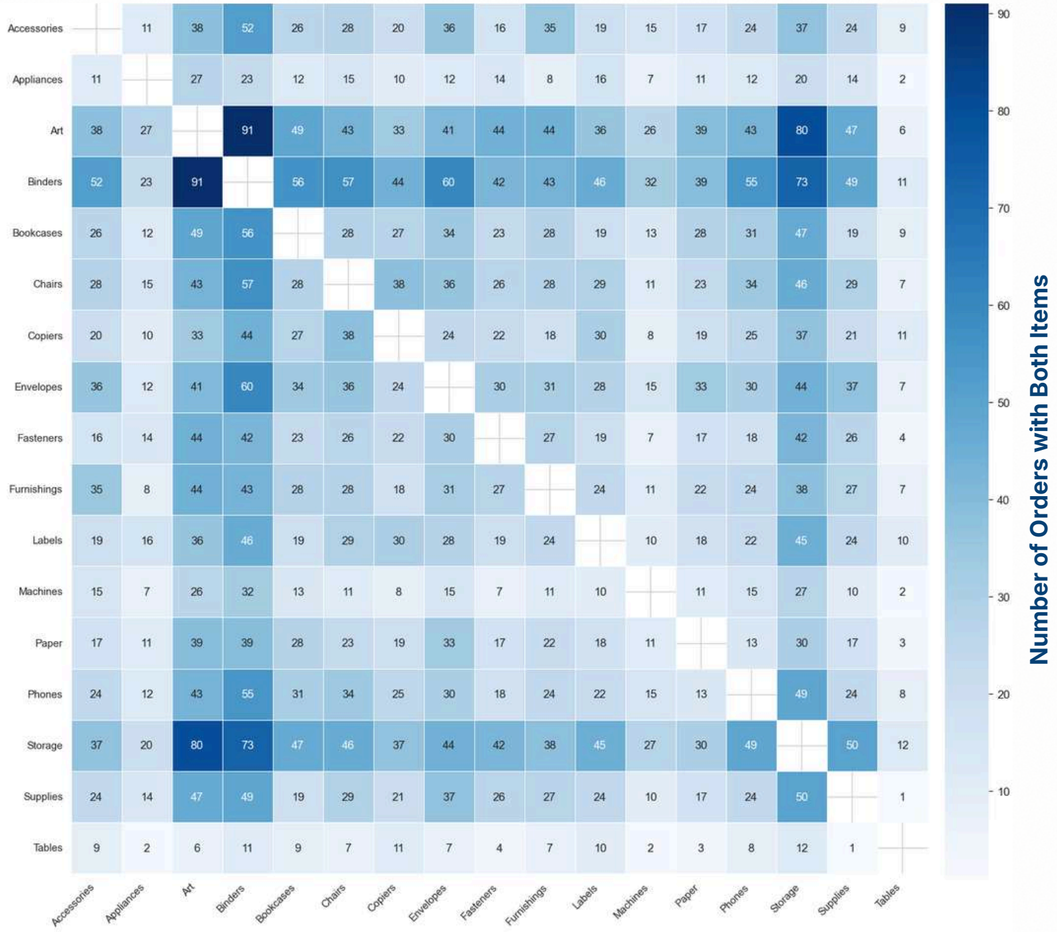
02. Predictive Analysis | Association

Sub-Category Co-occurence Heatmap



Angka di dalam kotak adalah jumlah transaksi riil (raw counts) di mana kedua barang dibeli bersamaan dalam satu pesanan.

Semakin gelap warna birunya, semakin tinggi frekuensi kedua barang tersebut muncul dalam keranjang yang sama.

Volume Transaksi Tertinggi: Pasangan Art & Binders adalah kombinasi mutlak paling laris dengan total 91 transaksi bersama. Ini ditandai dengan kotak biru paling pekat di seluruh matriks.

"Trio" Keranjang Utama: Kategori Art, Binders, dan Storage adalah motor penggerak keranjang belanja. Selain Art & Binders (91), pelanggan juga sangat sering membeli Art & Storage (80 transaksi) serta Binders & Storage (73 transaksi).

Kategori Independen: Tables (Meja) sangat jarang dibeli bersamaan dengan kategori lain. Angka persilangannya sangat rendah (contoh: dibeli bersama Supplies hanya 1 kali, dan bersama Appliances hanya 2 kali).

02. Predictive Analysis | Association

Association Lift Matrix

	Accessories	Appliances	Art	Binders	Bookcases	Chairs	Copiers	Envelopes	Fasteners	Furnishings	Labels	Machines	Paper	Phones	Storage	Supplies	Tables
Accessories		0.83	0.91	1.10	1.04	0.91	0.81	1.35	0.67	1.44	0.82	1.23	0.80	0.93	0.84	1.08	1.36
Appliances	0.83		1.12	0.85	0.83	0.85	0.71	0.78	1.02	0.57	1.20	1.00	0.90	0.81	0.79	1.09	0.52
Art	0.91	1.12		1.06	1.08	0.77	0.74	0.84	1.01	0.99	0.85	1.17	1.01	0.91	1.00	1.16	0.50
Binders	1.10	0.85	1.06		1.09	0.90	0.87	1.10	0.86	0.86	0.97	1.28	0.89	1.03	0.81	1.07	0.81
Bookcases	1.04	0.83	1.08	1.09		0.84	1.01	1.17	0.89	1.06	0.76	0.98	1.21	1.10	0.99	0.79	1.25
Chairs	0.91	0.85	0.77	0.90	0.84		1.15	1.01	0.81	0.86	0.94	0.68	0.81	0.98	0.78	0.97	0.79
Copiers	0.81	0.71	0.74	0.87	1.01	1.15		0.84	0.86	0.69	1.21	0.62	0.84	0.90	0.79	0.88	1.55
Envelopes	1.35	0.78	0.84	1.10	1.17	1.01	0.84		1.08	1.10	1.04	1.06	1.34	1.00	0.86	1.43	0.91
Fasteners	0.67	1.02	1.01	0.86	0.89	0.81	0.86	1.08		1.07	0.79	0.55	0.77	0.67	0.92	1.12	0.58
Furnishings	1.44	0.57	0.99	0.86	1.06	0.86	0.69	1.10	1.07		0.98	0.86	0.98	0.88	0.82	1.15	1.00
Labels	0.82	1.20	0.85	0.97	0.76	0.94	1.21	1.04	0.79	0.98		0.82	0.84	0.85	1.02	1.07	1.50
Machines	1.23	1.00	1.17	1.28	0.98	0.68	0.62	1.06	0.55	0.86	0.82		0.98	1.10	1.16	0.85	0.57
Paper	0.80	0.90	1.01	0.89	1.21	0.81	0.84	1.34	0.77	0.98	0.84	0.98		0.54	0.74	0.83	0.49
Phones	0.93	0.81	0.91	1.03	1.10	0.98	0.90	1.00	0.67	0.88	0.85	1.10	0.54		0.99	0.96	1.07
Storage	0.84	0.79	1.00	0.81	0.99	0.78	0.79	0.86	0.92	0.82	1.02	1.16	0.74	0.99		1.18	0.95
Supplies	1.08	1.09	1.16	1.07	0.79	0.97	0.88	1.43	1.12	1.15	1.07	0.85	0.83	0.96	1.18		0.16
Tables	1.36	0.52	0.50	0.81	1.25	0.79	1.55	0.91	0.58	1.00	1.50	0.57	0.49	1.07	0.95	0.16	

Lift (>1 = positive association): 1.6, 1.4, 1.2, 1.0, 0.8

Warna Hijau (Lift > 1.0):
Asosiasi positif. Pembelian barang A mendorong pembelian barang B. Semakin gelap hijaunya, hubungannya semakin kuat.

Warna Oranye/Merah (Lift < 1.0):
Asosiasi negatif. Pembelian barang A justru menurunkan kemungkinan barang B dibeli (biasanya karena barang tersebut saling menggantikan atau beda segmen).

Korelasi Positif Terkuat (Hijau Pekat):
Mengonfirmasi temuan sebelumnya, **Tables & Copiers (1.55), Tables & Labels (1.50), dan Accessories & Furnishings (1.44)** adalah pasangan emas yang sangat saling melengkapi.

Korelasi Negatif Terkuat (Merah/Oranye Pekat):

Pasangan **Tables & Supplies** memiliki nilai Lift terendah di seluruh matriks, yaitu **0.16**. Artinya, orang yang membeli meja hampir dipastikan tidakmembeli supplies di pesanan yang sama. Pasangan lain yang sangat tidak nyambung adalah **Tables & Paper (0.49) dan Paper & Phones (0.54).**

02. Predictive Analysis | Regression

00. Import Libraries

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

01. Menentukan variabel independen dan dependen

```
# 1. Select Features (X) and Target (y)
X = df[['Sales', 'Quantity', 'Discount', 'Shipping Cost']]
y = df['Profit']
```

02. Split data into *training and testing sets*

```
# 2. Split the data into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

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02. Predictive Analysis | Regression

03. Initialize and train the Linear Regression model

```
# 3. Initialize and train the Linear Regression model
regressor = LinearRegression()
regressor.fit(X_train, y_train)
```

04. Make predictions on the test set

```
# 4. Make predictions on the test set
y_pred = regressor.predict(X_test)
```

05. Evaluate the model

```
# 5. Evaluate the model
print(f"R-squared: {r2_score(y_test, y_pred):.4f}")
print(f"Mean Squared Error: {mean_squared_error(y_test, y_pred):.4f}")
```

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02. Predictive Analysis | Regression

06. View the Coefficients to see which factor impacts profit the most

```
# 6. View the Coefficients to see which factor impacts profit the most
coefficients = pd.DataFrame(regressor.coef_, X.columns, columns=['Coefficient'])
display(coefficients)
```

07. Plotting the graph (Actual Profit vs Predicted)

```
# Plot actual vs predicted values
sns.regplot(x=y_test, y=y_pred,
            scatter_kws={'alpha':0.4, 'color':'blue'},
            line_kws={'color':'red', 'lw':2})

plt.title('Regression Results: Actual vs. Predicted Profit', fontsize=14, fontweight='bold')
plt.xlabel('Actual Profit ($)', fontsize=12)
plt.ylabel('Predicted Profit ($)', fontsize=12)
```

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02. Predictive Analysis | Regression

08. Add reference line and print the graph

```
# Add a reference line for perfect predictions
max_val = max(y_test.max(), y_pred.max())
min_val = min(y_test.min(), y_pred.min())
plt.plot([min_val, max_val], [min_val, max_val], color='black', linestyle='--', label='Perfect Prediction Line')

plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print("\nHow to read this graph:")
print(" - The BLUE DOTS represent individual orders.")
print(" - The BLACK DASHED LINE is where dots would be if the model predicted profit perfectly.")
print(" - The RED LINE is the actual trend of our model.")
print(" - If dots are scattered far away from the lines, it means other hidden factors (not included in our data) are affecting the Profit.")
```

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02. Predictive Analysis | Regression

The Code:

```
print("--- REGRESSION ANALYSIS INTERPRETATION ---\n")

# Explain R-Squared
r2 = r2_score(y_test, y_pred)
print(f"Model Accuracy (R-squared): {r2:.2f}")
print(f"Explanation: The features we selected explain {r2*100:.1f}% of the variations in Profit. (100% would be a perfect prediction).\n")

print("Impact of each factor on Profit:")
# Loop through features and their corresponding coefficients
for feature, coef in zip(X.columns, regressor.coef_):
    direction = "INCREASES" if coef > 0 else "DECREASES"
    print(f"  - {feature}: For every 1 unit increase in {feature}, the Profit {direction} by ${abs(coef):.2f}.")

print("\nBusiness Insight:")
print("Look for the variables that decrease profit (usually Discount or Shipping Cost). These are areas where the business is bleeding money. Variables with high positive numbers are your strongest profit drivers.\n")

# --- PLOTTING THE GRAPH ---
plt.figure(figsize=(10, 6))

# Plot actual vs predicted values
sns.regplot(x=y_test, y=y_pred,
            scatter_kws={'alpha':0.4, 'color':'blue'},
            line_kws={'color':'red', 'lw':2})

plt.title('Regression Results: Actual vs. Predicted Profit', fontsize=14, fontweight='bold')
plt.xlabel('Actual Profit ($)', fontsize=12)
plt.ylabel('Predicted Profit ($)', fontsize=12)

# Add a reference line for perfect predictions
max_val = max(y_test.max(), y_pred.max())
min_val = min(y_test.min(), y_pred.min())
plt.plot([min_val, max_val], [min_val, max_val], color='black', linestyle='--', label='Perfect Prediction Line')

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print(" - If dots are scattered far away from the lines, it means other hidden factors (not included in our data) are affecting the Profit.")
```

Kelompok 1 & 10 | *Analitika Data Manajemen*
Carmen Kiara Indraputra (2306223856)

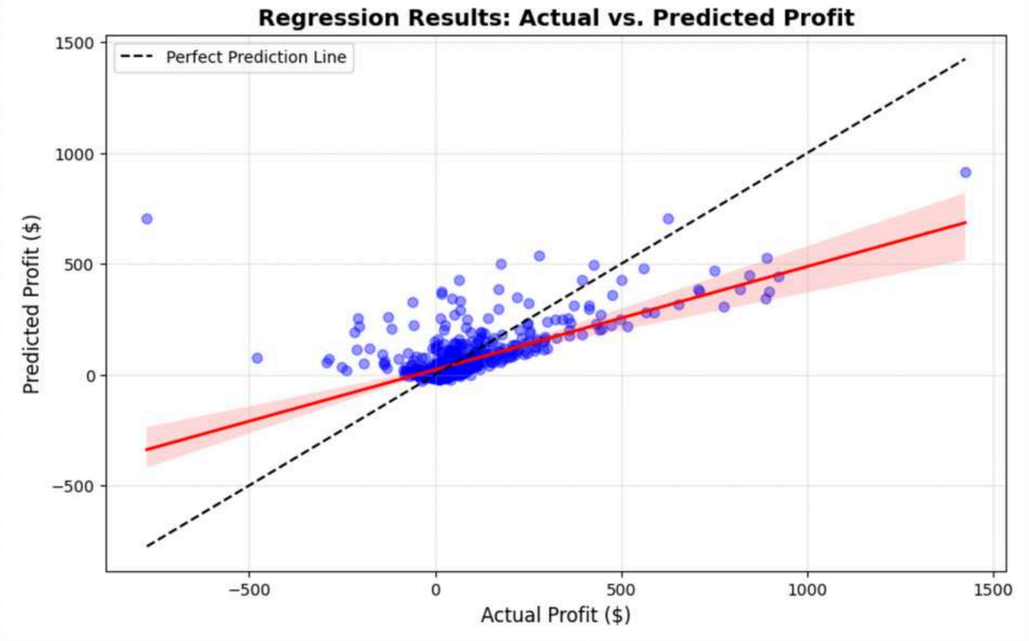
02. Predictive Analysis | Regression

Hasil:

R-squared: 0.4424

Mean Squared Error: 10488.1250

	Coefficient
Sales	0.223948
Quantity	-3.458132
Discount	0.000000
Shipping Cost	-0.123946

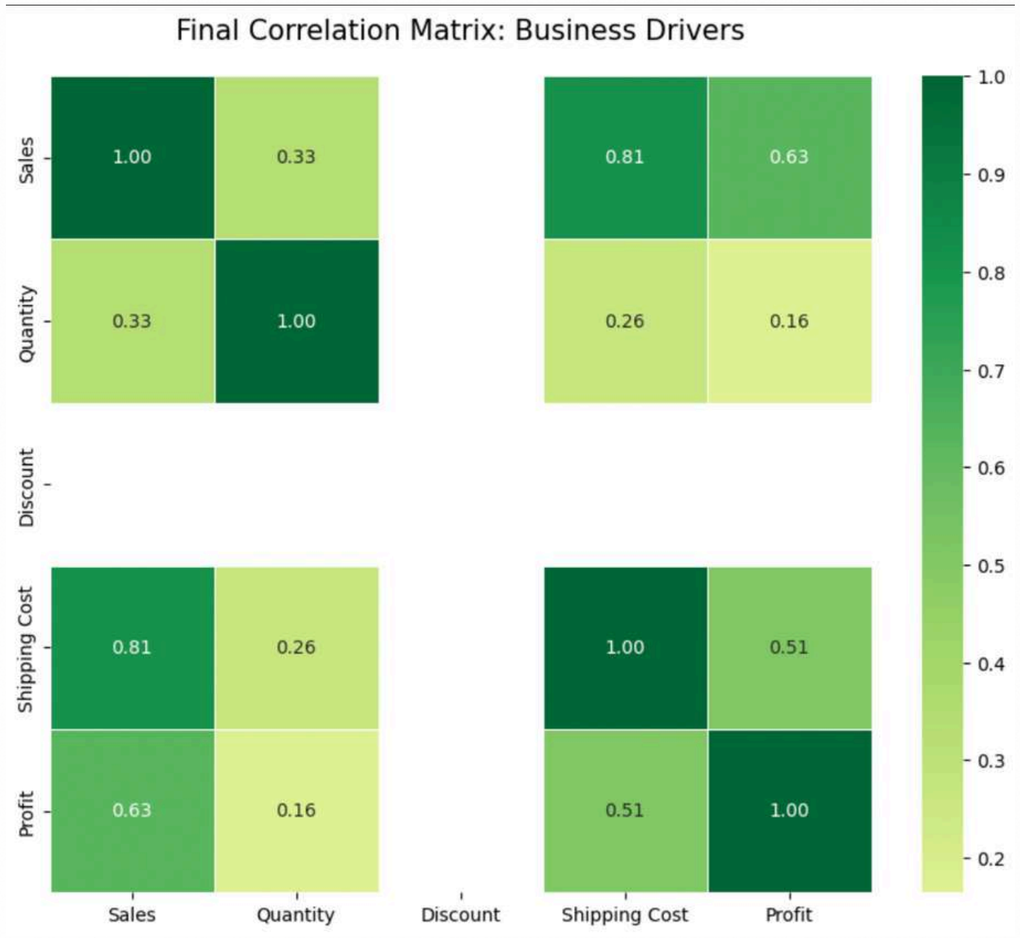


Individual Orders
Actual Profit
Predicted Profit

- Variabel sales, qty, discount, dan shipping cost dapat menjelaskan **44.2%** dari variasi profit
- Setiap kenaikan 1 unit Sales, Profit akan meningkat $0.22
- Setap kenaikan 1 unit Quantity, Profit akan berkurang $3.46 → **Terdapat variabel yang tidak dicantumkan mengenai operational efficiency perusahaan yang dapat menjelaskan kenaikan qty yang justru menurunkan profit**
- Setiap kenaikan 1 unit Discount, Profit akan berkurang $0
- Setiap kenaikan 1 unit Shipping Cost, Profit akan berkurang $1.23

Kelompok 1 & 10 | *Analitika Data Manajemen*
Carmen Kiara Indraputra (2306223856)

02. Predictive Analysis | Regression



Business Driver Correlation

1	Sales & Profit	Moderate POSITIVE Correlation
2	Quantity & Sales	Moderate POSITIVE relationship
3	Shipping Cost & Sales	Strong POSITIVE relationship
4	Shipping Cost & Profit	Moderate POSITIVE relationship

Revenue growth is successfully translating into bottom-line profit.

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Terima kasih